

Pakistan Crop Fertilizer Guide: NPK Recommendations and Application Timing

This guide provides fertilizer (NPK and micronutrient) recommendations for major crops grown in Pakistan, intended as a reference for farmer advisory use. Recommendations are general guidelines; actual doses should ideally be adjusted based on soil testing, since soil fertility varies significantly across Pakistan's provinces and even between fields in the same district.

General Fertilizer Principles for Pakistani Soils

Nitrogen (N), Phosphorus (P_2O_5), and Potassium (K_2O) are the three primary nutrients referenced in all crop fertilizer recommendations. These are typically supplied through Urea (46 percent nitrogen), Diammonium Phosphate or DAP (18 percent nitrogen, 46 percent phosphorus), Single Super Phosphate or SSP (16 percent phosphorus), and Sulphate of Potash (SOP) or Muriate of Potash (MOP), both providing 60 percent potassium. Nitrogen is prone to leaching and volatilization losses, so it is typically split into two or three applications across the crop's growth cycle rather than applied all at once. Phosphorus and potassium are far less mobile in soil and are typically applied in full as a single basal dose at sowing or land preparation, since delayed application reduces their effectiveness once the crop's root system has already developed past the early establishment stage.

Zinc deficiency is widespread across Pakistan's calcareous, alkaline soils, particularly affecting rice and wheat grown in the same rotation in central Punjab and parts of Sindh. Zinc deficiency in rice is locally known as khaira disease and causes bronzing and stunting of young seedlings; it is corrected with a basal application of zinc sulphate at approximately 25 kilograms per hectare. Boron deficiency is common in cotton and fruit orchards, corrected with borax application or foliar boron sprays. Iron deficiency causing leaf chlorosis (yellowing between leaf veins while veins remain green) is common in citrus grown on calcareous, high-pH soils and is managed with foliar iron chelate sprays, since soil-applied iron is often rendered unavailable by high soil pH before the plant can absorb it.

Wheat Fertilizer Recommendation

The general recommended fertilizer dose for irrigated wheat in Pakistan is approximately 120 to 150 kilograms of nitrogen, 85 to 115 kilograms of phosphorus (P_2O_5), and 60 kilograms of potassium (K_2O) per hectare, equivalent to roughly 50 to 60 kilograms of nitrogen, 35 to 45 kilograms of phosphorus, and 25 kilograms of potassium per acre. Full phosphorus and potassium, along with one-third to one-half of the total nitrogen dose, should be applied as a basal application at sowing. The remaining nitrogen is split between the first irrigation (at the Crown Root Initiation stage, approximately 20 to 25 days after sowing) and the second irrigation (at the tillering stage, approximately 40 to 45 days after sowing). On zinc-deficient soils, a basal application of 25 kilograms of zinc sulphate per hectare is recommended alongside the standard NPK dose.

Rice Fertilizer Recommendation

For Basmati rice, the recommended fertilizer dose is approximately 80 to 100 kilograms of nitrogen, 60 kilograms of phosphorus, and 60 kilograms of potassium per hectare, equivalent to roughly 35 to 45 kilograms of nitrogen, 25 kilograms of phosphorus, and 25 kilograms of potassium per acre. High-yielding coarse rice varieties require a higher nitrogen dose, up to 120 to 140 kilograms of nitrogen per hectare, reflecting their greater yield potential and biomass production. Full phosphorus and potassium, along with approximately one-third of the nitrogen dose, are incorporated into the soil during final puddling before transplanting. The remaining nitrogen is split between the active tillering stage and the panicle initiation stage. Zinc sulphate at 25 kilograms per hectare is commonly applied as a basal dose on zinc-deficient rice soils to prevent khaira disease.

Maize Fertilizer Recommendation

Hybrid maize is a heavy nutrient feeder, requiring approximately 200 to 250 kilograms of nitrogen, 100 to 125 kilograms of phosphorus, and 60 to 75 kilograms of potassium per hectare, equivalent to roughly 80 to 100 kilograms of nitrogen, 40 to 50 kilograms of phosphorus, and 25 to 30 kilograms of potassium per acre. This nitrogen requirement is considerably higher than wheat's, reflecting hybrid maize's greater yield potential and faster biomass accumulation. Phosphorus and potassium, along with approximately one-quarter of the total nitrogen dose, are applied basally at sowing. The remaining nitrogen is split into two to three top-dressing applications at the knee-high vegetative stage, the pre-tasseling stage, and the silking stage, since nitrogen availability during tasseling and silking has the strongest influence on final kernel set and yield.

Cotton Fertilizer Recommendation

Cotton fertilizer recommendations in Pakistan typically range from 100 to 150 kilograms of nitrogen, 60 to 90 kilograms of phosphorus, and 50 to 60 kilograms of potassium per hectare, adjusted according to variety and expected yield potential (hybrid Bt cotton varieties generally require the higher end of this range due to their greater yield potential and vegetative vigor). Phosphorus and potassium are applied basally at sowing, while nitrogen is split into three to four applications timed at sowing, squaring (bud formation), flowering, and boll development, since cotton has an extended fruiting period during which nutrient demand remains high. Boron deficiency, which causes flower and boll shedding, is a recognized concern in Pakistani cotton and is addressed with foliar boron sprays during the flowering and boll development period.

Potato Fertilizer Recommendation

Potato requires approximately 150 to 180 kilograms of nitrogen, 100 to 125 kilograms of phosphorus, and 125 to 150 kilograms of potassium per hectare, a notably higher potassium requirement compared to cereal crops, reflecting potassium's central role in tuber bulking and starch accumulation. Full phosphorus and potassium, along with part of the nitrogen dose, are applied basally at planting and ridging. The remaining nitrogen is applied at the earthing-up stage, approximately 25 to 30 days after planting, when ridges are built up around the growing plants both to supply nutrients for the tuber bulking phase and to prevent tuber greening from sunlight exposure.

Tomato Fertilizer Recommendation

Tomato fertilizer recommendations are approximately 120 to 150 kilograms of nitrogen, 80 to 100 kilograms of phosphorus, and 100 to 120 kilograms of potassium per hectare, with potassium being particularly important for fruit quality, firmness, and shelf life. Calcium availability, while not always a true soil deficiency, is closely tied to irrigation consistency in tomato; irregular watering is the classic cause of blossom-end rot even when soil calcium levels are otherwise adequate, since calcium uptake depends on steady water movement through the plant.

Chickpea (Gram) Fertilizer Recommendation

Chickpea, being a leguminous crop capable of biological nitrogen fixation through root nodule bacteria (*Rhizobium*), requires only a small starter dose of nitrogen, typically 15 to 20 kilograms per hectare, applied basally to support early seedling growth before the nitrogen-fixing nodules become fully active. Phosphorus requirement is higher, typically 60 to 75 kilograms per hectare, since phosphorus supports both root development and nodulation efficiency. Potassium is generally required only on potassium-deficient soils, at approximately 30 to 40 kilograms per hectare. Seed inoculation with *Rhizobium* culture before sowing is a recommended practice on fields where chickpea has not been grown recently, to ensure effective nodulation and reduce dependency on applied nitrogen fertilizer.

Sugarcane Fertilizer Recommendation

Sugarcane, being a long-duration crop occupying the field for ten to twelve months or more, requires a substantial fertilizer dose of approximately 200 to 250 kilograms of nitrogen, 100 to 125 kilograms of phosphorus, and 100 to 125 kilograms of potassium per hectare. Phosphorus and potassium, along with a portion of the nitrogen, are applied basally at planting, with the remaining nitrogen split across two to three applications during the crop's vegetative growth period, extending through the tillering and grand growth phases, since sugarcane's extended growth duration means nutrient demand continues over a much longer window than annual cereal crops.

Onion Fertilizer Recommendation

Onion fertilizer recommendations are approximately 100 to 120 kilograms of nitrogen, 60 to 80 kilograms of phosphorus, and 60 to 80 kilograms of potassium per hectare. Phosphorus and potassium are applied basally, while nitrogen is split into two to three applications, with the final nitrogen application timed before bulb initiation begins, since excessive late-season nitrogen application after bulb formation has started can delay maturity and reduce storage quality of the harvested bulbs.